

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science **ASSESSMENT**

TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE-AI	Date:	

Code of Conduct

I ____ T Jyothi Reddy (192472079)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ T Jyothi Reddy_____

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2

Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings son's account."	TRANSFER(SourceAccount, Amount, account to my DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules. **Field**

Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$
- $\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$
- $\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$
- $\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$
- $\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow$

PriorityIrrigation(x) **Tasks:**

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: "**Safari download for Windows**" and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	4	4	4	4	3	19	97
2	4	4	4	4	4	20	
3	5	4	5	4	4	22	
4	4	4	5	4	3	20	

Faculty In-charge	Course Coordinator	Program Director

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Frame Semantics in Banking Customer-Service Systems

1. Analyze the semantic frame of each banking request

a) **Transfer ₹20,000 from my savings account**

to my son's account Semantic frame:

TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)

- **Action:** Transfer
- **Source Account:** Savings account
- **Amount:** ₹20,000
- **Destination Account:** Son's account
- **Customer:** Account holder

b) **Block my debit card immediately** Semantic

frame:

BLOCK(DebitCard, Customer)

- **Action:** Block
- **Object:** Debit card
- **Customer:** Account holder

c) **Send my transaction statement to my**

email Semantic frame:

SEND(TransactionStatement, Email, Customer)

- **Action:** Send
- **Object:** Transaction statement
- **Destination:** Email
- **Customer:** Account holder

d) **Increase my daily withdrawal limit**

Semantic frame:

MODIFY(WithdrawalLimit, Customer)

- **Action:** Increase/Modify
- **Object:** Daily withdrawal limit
- **Customer:** Account holder

These frames identify the action and the participants involved in each banking request.

2. Identify the query in which the semantic interpretation is incorrect Query

B4 is incorrect.

- **Actual user intent:** Increase withdrawal limit
- **System interpretation:** Decrease withdrawal limit

The interpretation is incorrect because **increase and decrease have opposite meanings**. The system would therefore perform an operation different from what the customer requested.

3. Effect of incorrect identification of semantic roles

Incorrect identification of semantic roles can lead to incorrect banking operations. For example:

- Incorrect source or destination can transfer money to the wrong account.
- Incorrect amount can cause an incorrect transaction.
- Incorrect action can cause an increase to be performed as a decrease.
- Incorrect object can result in the wrong account, card, or service being modified.

Therefore, accurate semantic-role identification is important for safe and reliable banking operations.

4. Recommended frame-based semantic analysis approach A reliable

banking conversational system should:

1. Identify the action in the customer's request.
2. Identify all relevant semantic roles, such as customer, object, source, destination, and amount.
3. Construct the corresponding semantic frame.
4. Check the meaning using the surrounding context.
5. Validate important banking details before executing the operation.
6. Ask the customer for confirmation before sensitive transactions.

This frame-based approach can reduce semantic interpretation errors and improve the reliability of banking conversational systems.

Q2. First-Order Predicate Calculus for Smart Agriculture

1. Represent the field observations using First-Order Predicate Calculus The

given field observations are:

- **F1:** $\text{Dry}(\text{F1}) \wedge \text{IrrigationAvailable}(\text{F1}) \wedge \text{Rice}(\text{F1})$
- **F2:** $\text{Wet}(\text{F2}) \wedge \text{IrrigationAvailable}(\text{F2}) \wedge \text{Wheat}(\text{F2})$
- **F3:** $\text{Dry}(\text{F3}) \wedge \text{IrrigationUnavailable}(\text{F3}) \wedge \text{Maize}(\text{F3})$
- **F4:** $\text{Wet}(\text{F4}) \wedge \text{IrrigationUnavailable}(\text{F4}) \wedge \text{Rice}(\text{F4})$

The given rules are:

- $\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$
- $\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$
- $\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$
- $\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$
- $\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)$

2. Determine which fields require irrigation For F1:

$\text{Dry}(\text{F1}) \wedge$

$\text{IrrigationAvailable}(\text{F1})$

Therefore: $\text{NeedsWater}(\text{F1})$

Then:

NeedsWater(F1) $\rightarrow$ Irrigate(F1) So,

F1 requires irrigation.

For **F2** and **F4**, both fields are wet:

Wet(x) $\rightarrow$ $\neg$ NeedsWater(x)

Therefore, **F2 and F4 do not require irrigation.**

For **F3**:

Dry(F3) indicates that it needs water, but:

IrrigationUnavailable(F3) $\rightarrow$ $\neg$ Irrigate(F3) Therefore,

F3 cannot be irrigated.

Final result: F1 is the only field that should be irrigated.

3. Should F1 and F3 be treated differently?

Yes, F1 and F3 should be treated differently.

Field	Soil Condition	Irrigation Availability	Decision
F1	Dry	Available	Irrigate
F3	Dry	Unavailable	Do not irrigate

Although both fields are dry, F1 has irrigation available, while F3 does not. Therefore, the automated controller must consider both **water requirement and irrigation availability**.

4. Is predicate logic alone sufficient for agricultural decision support?

No.

Predicate logic is useful when sensor information is complete and certain. However, agricultural sensors may provide incomplete or uncertain information. Therefore, predicate logic alone may not always be sufficient. For better decision support, predicate logic can be combined with methods that handle uncertainty, such as probabilistic or fuzzy reasoning.

Q3. Word Sense Disambiguation in a Travel Recommendation System

1. Determine the intended sense of the ambiguous terms

The intended meanings can be determined from the query context and the user's interactions.

Search Query	Intended Sense	Supporting Context
Amazon tour	Amazon rainforest	User viewed rainforest trekking packages
Safari booking	Wildlife tour	User selected wildlife resort packages
Java vacation	Java island	User viewed hotels in Bali and Java
Cruise package	Ship journey	User viewed Mediterranean ship packages
Safari download for Windows	Web browser	User clicked a browser download page

Thus, the intended sense changes according to the **query context and user behaviour**.

2. Identify the lexical and contextual cues

- **Amazon + tour + trekking** $\rightarrow$ indicates the **Amazon rainforest**.
- **Safari + booking + wildlife resort** $\rightarrow$ indicates a **wildlife tour**.

- **Java + vacation + hotels + Bali** → indicates **Java island**.
- **Cruise + Mediterranean + ship packages** → indicates a **ship journey**.
- **Safari + download + Windows + browser page** → indicates the **Safari web browser**.

These surrounding words and user interactions help distinguish between the possible meanings.

3. Why may the same word require different interpretations for different users?

An ambiguous word can have multiple meanings. The correct meaning depends on:

- **Surrounding words and query context**
- **User search history**
- **User interests and previous interactions**
- **Pages or packages selected by the user**

For example, “Safari” means a wildlife tour in “Safari booking”, but refers to a web browser in “Safari download for Windows.” Therefore, the system should consider the user's context instead of assigning one fixed meaning to a word.

4. Industrial-scale WSD strategy

A large-scale travel recommendation system can use the following approach:

1. **Query Context:** Analyze surrounding words to identify possible meanings.
2. **User History:** Use previous searches and viewed content to understand user intent.
3. **Word Embeddings:** Represent words based on their contextual meaning.
4. **Click Behaviour:** Use clicked pages and selected packages as evidence of the intended sense.
5. **Sense Prediction:** Combine these features to select the most likely word sense.
6. **Recommendation:** Provide travel recommendations based on the selected meaning.

This approach combines query context, user history, embeddings, and click behaviour to improve Word Sense Disambiguation and travel recommendations.

Q4. Syntax-Driven Semantic Analysis in Legal Document Processing

1. Analyze how syntactic structure influences semantic-role assignment

Syntactic structure helps the NLP system identify the roles of entities in a sentence, such as **Agent, Object, and Recipient**.

Sentence 1 – Active voice:

“The supplier delivered the equipment to the buyer.”

- **Supplier** → **Agent**
- **Equipment** → **Object**
- **Buyer** → **Recipient**

Here, the grammatical subject **supplier** is also the Agent.

Sentence 2:

“The buyer received the equipment from the supplier.” The roles should be:

- **Buyer** → **Recipient**
- **Equipment** → **Object**
- **Supplier** → **Agent/Source**

Sentence 3 – Active voice:

“The company terminated the contract after the violation.”

- **Company → Agent**
- **Contract → Object**

Sentence 4 – Passive voice:

“The contract was terminated by the company.”

- **Contract → Object/Patient**
- **Company → Agent**

Thus, in passive constructions, the grammatical subject may be the **Object/Patient**, not the Agent.

2. Identify the incorrect semantic-role assignments There are **two incorrect interpretations**.

Sentence 2:

“The buyer received the equipment from the supplier.”

Given incorrect interpretation:

- Buyer = Agent
- Equipment = Object
- Supplier = Recipient **Correct interpretation:**
- Buyer = **Recipient**
- Equipment = **Object**
- Supplier = **Agent/Source** **Sentence 4:**

“The contract was terminated by the company.” Given

incorrect interpretation:

- Contract = Agent
- Company = Object **Correct interpretation:**
- Contract = **Object/Patient**
- Company = **Agent**

The errors occur because the system incorrectly assigns semantic roles based only on the grammatical structure.

3. Effect of confusing grammatical subject with semantic Agent

A grammatical subject is **not always the semantic Agent**, especially in passive sentences.

For example:

“The contract was terminated by the company.”

The **contract** is the grammatical subject, but it does not perform the action.

- Contract → **Object/Patient**
- Company → **Agent**

If an NLP system treats every grammatical subject as the Agent, it may:

- Assign responsibility to the wrong entity.
- Extract incorrect legal relationships.
- Identify the wrong party as performing an action.
- Produce incorrect legal information.

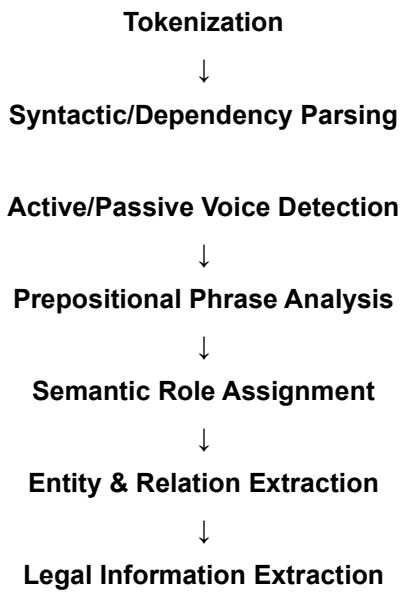
Therefore, confusing grammatical subjects with semantic Agents can seriously affect automated legal information extraction.

4. Recommended syntax-driven semantic analysis architecture A

suitable architecture for legal NLP is:

Legal Sentence





The system should be designed to correctly handle:

1. **Active voice**
2. **Passive voice**
3. **Prepositional phrases**
4. **Long-distance dependencies**
5. **Agent, Object, and Recipient roles**

This syntax-driven approach helps the system correctly connect syntactic structures with semantic roles and improves the accuracy of legal information extraction.